Modelado de Datos Autónomo: Sistemas Multi-Agente para Normalización 3NF y Generación de Esquemas

Autonomous Data Modeling: Multi-Agent Systems for 3NF Normalization and Schema Generation

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—agentes autónomos, sistemas multi-agente, normalización de bases de datos, tercera forma normal (3FN), generación de esquemas, diccionario de datos, modelos de lenguaje grandes (LLM), PostgreSQL.**

I. Introducción

El diseño de bases de datos relacionales consiste en traducir requisitos de negocio, habitualmente expresados en lenguaje natural, en un esquema físico capaz de soportar la operación de un sistema con garantías de integridad, consistencia y desempeño. Los cimientos de este proceso fueron establecidos por Codd con el modelo relacional [1] y consolidados mediante la definición formal de las dependencias funcionales y de las formas normales [2], mientras que el modelo entidad-relación [3] aportó el puente semántico entre el mundo real y la estructura lógica. Bajo este marco, alcanzar la Tercera Forma Normal (3FN) no es un objetivo cosmético: supone la eliminación de las anomalías de inserción, actualización y borrado, y permite construir diccionarios de datos coherentes y trazables.

No obstante, la práctica profesional dista mucho de este ideal. Desde la toma de requisitos hasta la redacción del código DDL, el diseñador debe identificar entidades y atributos, determinar dependencias funcionales, detectar dependencias transitivas, descomponer relaciones y documentar cada columna en un diccionario de datos. Cada una de estas etapas es vulnerable al error humano, y las herramientas CASE comerciales, como TOAD Data Modeler, asisten la representación gráfica y la generación parcial de código, pero no automatizan la decisión semántica que garantiza la normalización. En especificaciones complejas —con decenas de entidades, claves compuestas y cadenas de dependencias transitivas—, la carga cognitiva y la presión de plazos producen esquemas que incumplen la 3FN, acumulan redundancia disfrazada de rendimiento y generan diccionarios incoherentes, cuyo costo se posterga a migraciones y correcciones en producción.

La aparición de los modelos de lenguaje grandes (LLM) ha abierto una vía prometedora para automatizar este ciclo de vida. Trabajos recientes han mostrado que los LLM pueden derivar modelos conceptuales y esquemas preliminares a partir de especificaciones en lenguaje natural [4], y que arquitecturas multi-agente son capaces de generar esquemas relacionales de alta calidad desde requisitos [5]. Al mismo tiempo, la literatura evidencia límites claros: Kohita [6] demostró que el nivel de normalización del esquema modifica de forma sistemática la precisión con que los LLM generan SQL; Kanchinadam *et al.* [7] mostraron que variantes estructurales del mismo modelo conceptual producen respuestas divergentes en tareas Text-to-SQL; y ERBench [8] evidenció que los LLM cometen errores lógicos al deducir relaciones entre entidades, incluso en esquemas sencillos. Estos resultados convergen en una conclusión: la generación confiable de esquemas no puede delegarse en un único prompt, sino que requiere una verificación explícita y continua de las dependencias funcionales.

En el ámbito de las arquitecturas multi-agente, la evidencia contemporánea confirma que la especialización de roles reduce los errores lógicos. AttributeForge [9] orquesta 43 agentes especializados para modelar esquemas de producto en un entorno

de producción, mientras que GalaxyWeaver [10] demuestra que un ecosistema de agentes puede evaluar iterativamente un esquema deficiente y reestructurarlo de forma autónoma hasta alcanzar un estado óptimo. Estos antecedentes sugieren que la misma estrategia puede aplicarse al problema de la normalización relacional, siempre que exista un agente dedicado a auditar las dependencias matemáticas que sostienen el esquema.

En este artículo se presenta un framework empírico de modelado de datos autónomo compuesto por agentes especializados: un agente analista extrae entidades, atributos y relaciones a partir del texto de requisitos; un agente generador construye el diccionario de datos; un agente codificador redacta el código DDL para PostgreSQL; y un agente orquestador evalúa las dependencias funcionales e itera sobre el esquema hasta demostrar su conformidad con la 3FN antes de emitir el DDL final. Las contribuciones del artículo son: (i) la definición de una arquitectura multi-agente con roles cognitivos separados para extracción de requisitos, documentación y codificación, auditada por un verificador de dependencias; (ii) un protocolo de verificación que opera bajo los criterios formales de normalización de Codd [2] como métrica de éxito; (iii) una evaluación empírica sobre especificaciones complejas reales que mide la tasa de éxito del sistema para producir esquemas en 3FN libres de anomalías; y (iv) una comparación cuantitativa del tiempo de diseño frente al proceso manual con TOAD Data Modeler.

El resto del artículo se organiza como sigue. La Sección II presenta el marco teórico de normalización sobre el que se apoya el sistema. La Sección III describe la arquitectura del framework y los protocolos de interacción entre agentes. La Sección IV detalla el protocolo experimental y los casos de prueba. La Sección V analiza los resultados obtenidos y la Sección VI resume las conclusiones.

II. Ease of Use

A. Maintaining the Integrity of the Specifications

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A. Abbreviations and Acronyms

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B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as “3.5-inch disk drive”.
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- Use a zero before decimal points: “0.25”, not “.25”. Use “cm^3”, not “cc”.)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{1}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use “(1)”, not “Eq. (1)” or “equation (1)”, except at the beginning of a sentence: “Equation (1) is . . .”

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Please use “soft” (e.g., `\eqref{Eq}`) cross references instead of “hard” references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

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E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
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- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [17].

F. Authors and Affiliations

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TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
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[a]Sample of a Table footnote.



Fig. 1. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an

example, write the quantity “Magnetization”, or “Magnetization, M”, not just “M”. If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write “Magnetization (A/m)” or “Magnetization {A[m(1)]}”, not just “A/m”. Do not label axes with a ratio of quantities and units. For example, write “Temperature (K)”, not “Temperature/K”.

ACKNOWLEDGMENT

The preferred spelling of the word “acknowledgment” in America is without an “e” after the “g”. Avoid the stilted expression “one of us (R. B. G.) thanks . . .”. Instead, try “R. B. G. thanks. . .”. Put sponsor acknowledgments in the unnumbered footnote on the first page.

REFERENCES

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Unless there are six authors or more give all authors’ names; do not use “et al.”. Papers that have not been published, even if they have been submitted for publication, should be cited as “unpublished” [14]. Papers that have been accepted for publication should be cited as “in press” [15]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [16].

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